
IFF-AR: Conditional Autoregressive Audio Generation via Instantaneous Frequency Modelling

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Abstract

This report presents **IFF-AR**, an autoregressive audio generation model in the frequency domain, based on the joint prediction of log-magnitude and instantaneous frequency (IF). The architecture combines a random convolutional encoder (TCN) with a magnitude decoding head *Mag-Head* and a conditioned *normalising flow* for phase modelling. The loss function acts on the three domains—spectral, phase, and temporal—ensuring perceptual and structural consistency. This formulation allows for consistent time-frequency generation without iterative phase reconstruction.

1. Introduction

In recent years, deep generative models have made remarkable progress in sound synthesis, although direct phase modelling remains a challenge. Most existing approaches reconstruct phase iteratively, for example using Griffin-Lim (Griffin & Lim, 1984). This project proposes **IFF-AR**, an autoregressive model that generates audio directly in the frequency domain by jointly predicting *log-magnitude* and *instant frequency* (Takamichi et al., 2018). The architecture integrates a temporal convolutional encoder, a lightweight decoder for magnitude, and a conditional normalising flow (Dinh et al., 2016) for phase modelling. The phase is then treated as a learnable probabilistic variable conditioned on spectral energy, avoiding the rigid separation between magnitude and phase. Experimental results show that the proposed formulation produces stable and consistent spectrograms, highlighting the potential of conditioned flows for explicit phase modelling.

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2. Related Works

Early autoregressive models operated in the time domain, such as **WaveNet** (Van Den Oord et al., 2016), which synthesizes audio sample-by-sample. In the spectral domain, models like **MelNet** (Vasquez & Lewis, 2019) showed that predicting spectrogram sequences autoregressively reduces temporal length and complexity. **GANSynth** (Engel et al., 2019) introduced the use of instantaneous frequency (IF) as a phase representation, demonstrating audio generation with GANs. Diffusion models redefine the state of the art: **DiffWave** (Kong et al., 2020) provides a non-autoregressive pipeline for waveform synthesis; **AudioLDM** (Liu et al., 2023) applies latent diffusion on CLAP embeddings; **FlashAudio** (Liu et al., 2024) uses rectified flows for efficient one-step generation. Autoregressive Transformers, e.g. **Next-Scale** (Qiu et al., 2024) and **SongGen** (Liu et al., 2025), focus on sequence compression and controllable generation. The phase has traditionally been handled with iterative algorithms like Griffin-Lim, which require many iterations and often produce artefacts. Neural approaches have sought to improve this: (Takamichi et al., 2018) model phase with a von Mises DNN, while (Masuyama et al., 2023) show that predicting phase differences between adjacent frames is more stable and suitable for causal reconstruction. However, few approaches integrate this representation into a conditional autoregressive framework, simultaneously predicting magnitude and IF. This is the main novelty of this project.

3. Method

IFF-AR model architecture consists of four main modules, which are necessary to perceive the general context, predict magnitude and phase. During preprocessing, the audio is converted into a time-frequency representation using Short-Time Fourier Transform (STFT). This transformation produces complex coefficients, from which it is possible to extract two sets of features: *Log-magnitude*, log-scaled spectral power, standardised over the training set; *Instantaneous frequency (IF)* calculated as the phase difference between consecutive frames computed per-frequency as $IF_t(f) = \Delta\phi_t(f)/\Delta t$, describing phase dynamics di-

rectly without absolute-angle ambiguity. IF is normalised using the median and median absolute deviation (MAD). Each example is split into context–target windows for autoregressive prediction. **TCNContextEncoder** is implemented as a 1D random convolutional network with dilations, inspired by WaveNet (Van Den Oord et al., 2016), but instead of working at the audio sample level it operates directly at the frame level. This choice allows long-range dependencies in the time-frequency domain to be modelled effectively, avoiding Transformers and keeping computation low through convolutional parallelism. The encoder receives the log-magnitude and instantaneous frequency sequence of the context as input and produces a compact embedding that will subsequently be used both by the MagHead module and in the normalising flow conditioned for phase modelling. **MagHead** is a decoding head that uses embeddings provided by the TCNContextEncoder to predict the log-magnitudes of STFT frames, normalised as in the preprocessing stage. It consists of a light sequence of fully-connected layers and 1D convolutions with small kernels that project the embedding onto the frequency dimension; the architecture is deliberately light to reduce overfitting and to focus the model’s capacity on the flow for the phase. Furthermore, MagHead not only produces magnitudes, but also provides a condition for flow, in this way, the phase (IF) is modelled consistently with spectral energy: this condition is commonly avoided in models that treat magnitude and phase separately (Nugraha et al., 2019; Dai et al., 2025). **IFConditionalFlow** uses a conditional normalising flow to learn the distribution of future IFs explicitly and consistently with the predicted magnitudes, similar to the RealNVP design (Dinh et al., 2016). Each layer divides the input vector into two parts, x_a and x_b , and calculates, through small conditional networks, the shift and scale by applying an invertible transformation shown in the appendix in equation 1. Each coupling layer receives the context embedding (from the TCN) and the predicted magnitude (from the MagHead) as conditioning; the transformations depend not only on the IF input but also on the context. This choice represents an innovative element with respect to the literature: the phase is not treated as a residual or reconstructed iteratively (Griffin & Lim, 1984), nor estimated separately from conditional networks only on magnitudes (Takamichi et al., 2018; Masuyama et al., 2023), but explicitly modelled as a conditional probability distribution. Finally, to complete the architecture, it is necessary to reconstruct the absolute phase of the STFT coefficients to recover the time-domain signal via: $\phi_t(f) = \phi_{t-1}(f) + \Delta\phi_t(f)$ where $\Delta\phi_t(f)$ is the predicted phase. Next, the complex coefficients of the spectrogram are reconstructed as: $X_t = \hat{M}_t(f) \cdot e^{i\phi_t(f)}$ where $\hat{M}_t(f)$ are the magnitudes predicted by MagHead. Finally, the Inverse Short-Time Fourier Transform (iSTFT) is applied to obtain the reconstructed audio signal: $\hat{x} = iSTFT\{X_t(f)\}$.

The loss function was designed to deal with three different domains: spectral, phase and time. Many architectures, such as WaveNet (Van Den Oord et al., 2016) and MelNet (Vasquez & Lewis, 2019), do not include perceptual or structural terms such as Scale-Invariant Signal-to-Distortion Ratio (SI-SDR) or Multi-Resolution Short-Time Fourier Transform Loss (MRSTFT), which enhance the perceptual quality and consistency of the generated spectrum. The total loss is a weighted combination of these terms and is optimized using the Adam optimizer with a learning rate of 10^{-3} and early stopping on the validation loss.

4. Experimental Results

The NSynth dataset (Engel et al., 2017), segmented into context windows L and target windows K , was used for training. STFT was applied with FFT = 2048, hop = 512, and Hann window. Training was performed on Leonardo HPC (Cineca) (CINECA Supercomputing Centre & Department, 2024) using teacher forcing, with autoregressive predictions packaged on the actual target sequences. The model was trained for 30 epochs using Adam optimizer with learning rate $3 \cdot 10^{-4}$, batch size 32 and teacher forcing for autoregressive rollouts. The evaluation was performed on 2500 unseen sample. Table 1 reports the main metrics computed on the test set. Log-magnitude errors (MAE and RMSE) show that the decoder effectively captures the energy structure of the signal, reproducing both the global and local spectral patterns. The instantaneous frequency metrics indicate that the conditional flow models the phase dynamics with stable precision within the normalised range $[-\pi, \pi]$. SI-SDR value confirms perceptually consistent reconstruction with limited artefacts, demonstrating that the joint magnitude-phase formulation preserves the temporal coherence of the waveform. The predicted log-magnitude spectrogram 1 reproduce the harmonic structure and overall energy distribution of the targets, maintaining a high degree of temporal coherence across frames. Minor smoothing of high-frequency components can be observed, but the model consistently reconstructs the main formants, harmonic peaks and amplitude envelopes, showing that the magnitude and phase estimation evolve jointly and coherently. Audio reconstructions further supports these findings. The generated signals exhibit stable phase behaviour and realistic timbre.

5. Conclusions

IFF-AR is an autoregressive model in the time-frequency domain that combines log-magnitude and instantaneous frequency prediction in a conditional framework. The introduction of a normalising flow for explicit phase modelling, conditioned on both context and spectral energy,

represents an innovative element compared to existing models. Results demonstrate stable spectral structures, realistic timbre, and temporal coherence, confirming the effectiveness of direct IF prediction for complex-valued generation.

6. Appendix

Evaluation metrics. This table reports the quantitative evaluation computed on the NSynth test set.

Metric	Value	Metric	Value
LogMag MAE	27.477	LogMag RMSE	33.358
IF MAE	1.640	IF RMSE	2.004
LSD (dB)	3.719	SI-SDR (dB)	-1.537
SI-SDR (mag GT)	-1.460		

Table 1. Quantitative evaluation on the NSynth test set.

Predicted vs target spectrogram. The predicted log-magnitude spectrogram reproduces the main energy structure of the target while smoothing high-frequency components.

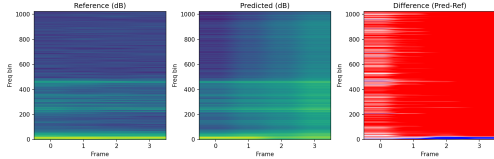


Figure 1. Predicted (center), target (left) log-magnitude spectrogram and difference (right).

Coupling transformation in IFConditionalFlow. Each coupling layer transforms part of the input (x_b) using scale and translation parameters (s, t) conditioned on both the complementary partition (x_a) and the context c . This formulation follows the RealNVP paradigm and ensures invertibility while allowing context-dependent modelling of phase dynamics.

$$\begin{cases} y_a = x_a \\ y_b = x_b \cdot \exp(s(x_a, c)) + t(x_a, c) \end{cases} \quad (1)$$

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